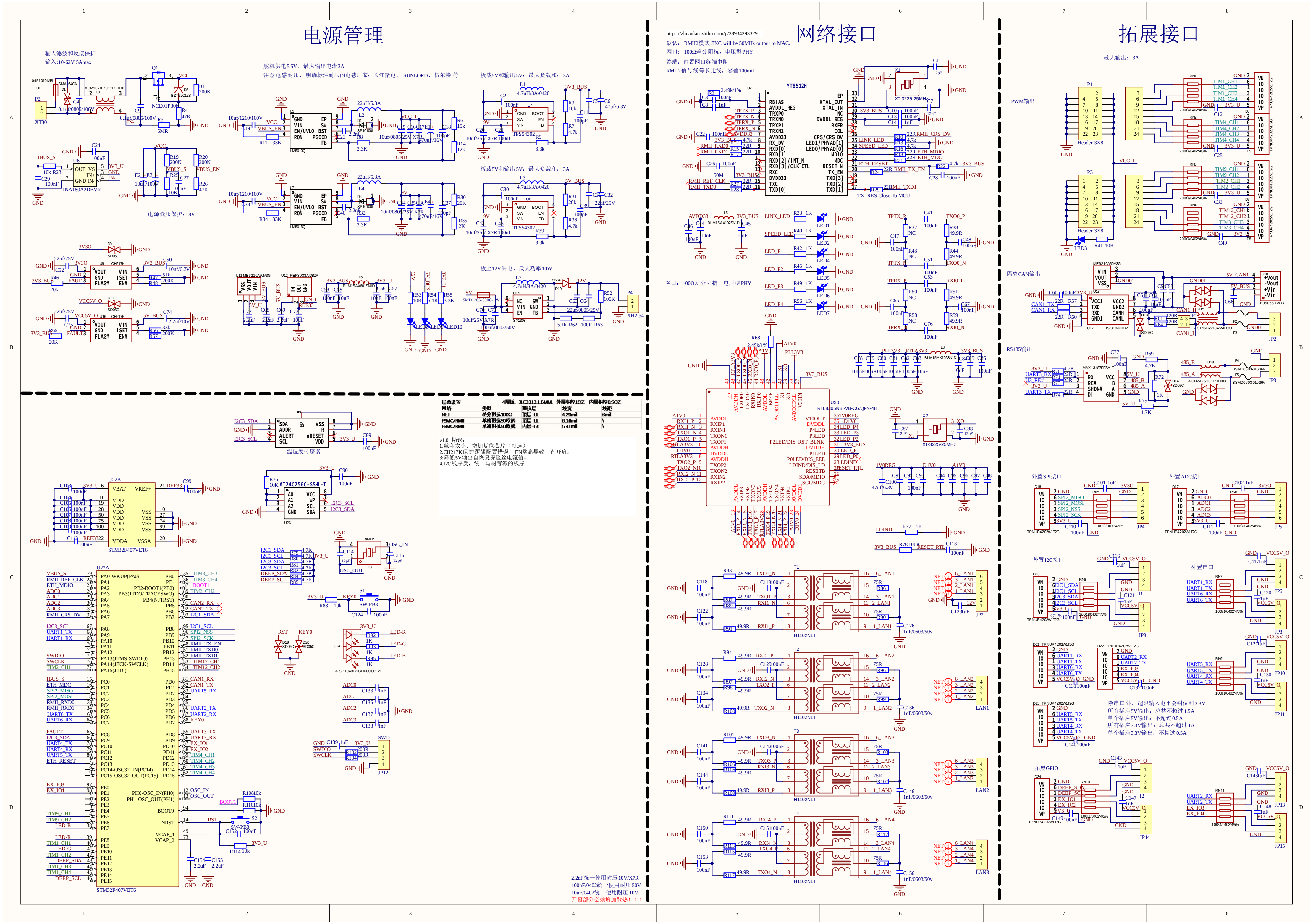
电源管理

网络接口

拓展接口



舵机供电5.5V，最大输出电流3A

注意电感耐压，明确标注耐压的电感厂家：长江微电， SUNLORD，伍尔特,等

板载5V和输出5V：最大负载和：3A

板载5V和输出5V：最大负载和：3A

板上12V供电，最大功率10W

https://zhuanlan.zhihu.com/p/28934293329

默认：RMII2模式:TXC will be 50MHz output to MAC.

网口：100Ω差分阻抗，电压型PHY

终端：内置网口终端电阻

RMII2信号线等长走线，容差100mil

层叠设置	4层板，JLC3313,1.6MM，外层铜厚1OZ，内层铜厚0.5OZ			
网络	类型	阻抗层	线宽	线距
NET	差分阻抗100Ω	顶层-L1	4.29mil	6mil
FSMC/RMII	单端阻抗50欧姆	顶层-L1	6.16mil	\
FSMC/RMII	单端阻抗50欧姆	内层-L3	5.41mil	\

v1.0 勘误：
1.丝印太小；增加复位芯片（可选）
2.CH217K保护逻辑配置错误，EN常高导致一直开启。
3:降低5V输出自恢复保险丝电流值。
4.I2C线序反，统一与树莓派的线序

除串口外，超限输入电平会钳位到3.3V
所有插座5V输出：总共不超过1.5A
单个插座5V输出：不超过0.5A
所有插座3.3V输出：总共不超过1A
单个插座3.3V输出：不超过0.5A

2.2uF统一使用耐压10V/X7R
100nF/0402统一使用耐压50V
10uF/0402统一使用耐压10V
开窗部分必须增加散热！！！

Comment	Description	Designator	Footprint	LibRef	Quantity
12pF	无极性贴片电容	C1, C7, C87, C88, C114, C115	C 0402_L	C	6
100nf	无极性贴片电容	C2, C3, C10, C12, C13, C21, C22, C24, C25, C26, C27, C28, C29, C30, C33, C41, C43, C46, C47, C48, C49, C51, C53, C54, C57, C58, C60, C61, C65, C67, C76, C77, C78, C79, C80, C81, C82, C85, C86, C89, C90, C91, C92, C93, C94, C95, C96, C97, C98, C99, C103, C104, C105, C106, C107, C108, C109, C110, C111, C112, C113, C118, C119, C122, C124, C125, C128, C129, C131, C132, C134, C140, C141, C142, C144, C149, C150, C151, C152, C153	C 0402_L	C	80
0.1uf/0805/100V	无极性贴片电容	C4, C9	C 0805_L	C	2
47uf/6.3V	无极性贴片电容	C5, C6, C100	C 0805_L	C	3
1uF	无极性贴片电容	C8, C14, C101, C102, C116, C117, C120, C121, C123, C127, C130, C139, C143, C145, C147, C148	C 0402_L	C	16
100pF	无极性贴片电容	C11, C18, C37, C39	C 0402_L	C	4
10uf/0805/25V X7R	无极性贴片电容	C15, C16, C17, C34, C35, C36	C 0805_L	C	6
10uf/1210/100V	无极性贴片电容	C19, C38	C 1210_L	C	2
10uf/25V X7R	无极性贴片电容	C20, C42, C70	C 0805_L	C	3
2.2nF	无极性贴片电容	C23, C40	C 0402_L	C	2
22uf/25V	无极性贴片电容	C31, C32, C52, C75	C 0805_L	C	4
10uF	无极性贴片电容	C44, C45, C56, C59, C62, C73, C83, C84	C 0402_L	C	8
10uf/6.3V	无极性贴片电容	C50	C 0402_L	C	1
22uf/10V X7R	无极性贴片电容	C55, C66	C 0805_L	C	2
22uf/0805/25V	无极性贴片电容	C63, C64	C 0805_L	C	2
2.2uF	无极性贴片电容	C68, C69, C72, C154, C155	C 0402_L	C	5
100nf/0603/50V	无极性贴片电容	C71	C 0603_L	C	1
2.2uf/10V	无极性贴片电容	C74	C 0402_L	C	1
1nF/0603/50v	无极性贴片电容	C126, C136, C146, C156	C 0603_L	C	4
1nF	无极性贴片电容	C133, C135, C137, C138	C 0402_L	C	4
SMAJ64CA		D1	SMA - duplicate	SMAJ33CA_C115251	1
BZT52C12S		D2	SOD-323 - duplicate	BZT52C12S_C19077447	1
TPNUP4202W1T2G		D3, D5, D7, D8, D16, D17, D18, D21, D22, D23, D24	SOT-363	NUP4202W1T2G_C2937014	11
SP10100L		D4, D6	TO277A	SB10100L_C113946	2
SD05C		D9, D11, D13, D14, D19, D20	SOD-323 - duplicate	SD05C	6
SS34		D10	SMA - duplicate	B5819W SL	1
PSM712-LF-T7		D12, D15	SOT23-3L - duplicate	PSM712-LF-T7	2
470uF/16V	直插固态电容	E1, E4	CM D(6.3*7.7)	C-CS	2
10uF/100V	直插固态电容	E2, E3	CM D(6.3*7.7)	C-CS	2
SMD1206-300C-12V		F1	F1206	BSMD1206-100-24V	1
BSMD0603-010-36V		F2, F3, F4, F5	F0603	SMD0603-050-16	4
HDR-1X4	4P接插件	I1, I2, JP6, JP8, JP9, JP10, JP11, JP13, JP14, JP15	GH1.25-WS-4P	Header 4	10
CAN_JUMP	2*2P接插件	JP1	HDR1.27-M-LI-2x2P	CAN_JUMP	1
GH1.25-WS-3P	3P接插件	JP2, JP3	GH1.25-WS-3P	Header 3	2
HDR-1X6	6P接插件	JP4, JP5	GH1.25-WS-6P	Header 6	2
CAM	6P接插件	JP7	MX 1.25-WS-6P	Header 6	1
SWD	4P接插件	JP12	HDR2.54-LI-4P	Header 4	1
4.7uH/3A/0420	Inductor	L1, L3, L7	L-SMD-0420	Inductor_6*6	3
22uH/5.3A	MWSA1005S-220MT	L2, L4	WE-HCI_1050	Inductor_6*6	2
BLM15AX102SN1D		L5, L8	L 0402_L	BLM15AX102SN1D	2
BLM15AX601SN1D		L6	L 0402_L	BLM15AX102SN1D	1
MX1.25	4P接插件	LAN1, LAN2, LAN3	MX 1.25-WS-4P	Header 4	3
绿色 0603	贴片LED	LED1, LED2, LED4, LED5, LED6, LED7	LED 0402	LED-SMD	6
绿色 0402	贴片LED	LED3, LED8, LED9, LED10	LED 0402	LED-SMD	4
Header 3X8		P1, P3	Header 3X8	Header 3X8	2
XT30	2P接插件	P2	XT30 PW-M	Header 2	1
XH2.54	2P接插件	P4	XH2.54-WI-2P	Header 2	1
NCE01P30K	MOS(场效应管)	Q1	TO252	AO3401A P道沟 X13A15 SOT-23	1
200K	贴片电阻	R1, R19, R20	R 0603_L	R	3
2.49k/1%	贴片电阻	R2, R68	R 0402_L	R	2
10k	贴片电阻	R3, R23, R25, R41, R53, R76, R88, R108, R110, R114	R 0402_L	R	10
47K	贴片电阻	R4	R 0603_L	R	1
5MR	贴片电阻	R5	R 0805_L	R	1
15k	贴片电阻	R6	R 0402_L	R	1
4.7k	贴片电阻, RES_0603	R7, R12, R13, R16, R22, R36, R69, R70, R75, R79, R80, R81, R82, R84, R85	R 0402_L	R, RES_0603	15
3.3K	RES_0603, 贴片电阻	R8, R9, R32, R39, R55	R 0402_L	RES_0603, R	5
22R	贴片电阻, RES_0603	R10, R15, R17, R18, R21, R24, R27, R28, R29, R57, R60, R71, R73, R74	R 0402_L	R, RES_0603	14
33K	RES_0603, 贴片电阻	R11, R34, R66	R 0402_L	RES_0603, R	3
12k	贴片电阻	R14	R 0402_L	R	1
47K	贴片电阻	R26	R 0402_L	R	1
20K	RES_0603, 贴片电阻	R30, R31, R46, R65	R 0402_L	RES_0603, R	4
1K	贴片电阻, RES_0603	R33, R40, R42, R45, R49, R56, R72, R77, R92, R93, R95	R 0402_L	R, RES_0603	11
2K	RES_0603	R35	R 0402_L	RES_0603	1
NC	贴片电阻	R37, R43, R50, R58	R 0402_L	R	4
49.9R	贴片电阻	R38, R44, R51, R59, R83, R87, R89, R91, R94, R97, R98, R100, R101, R105, R106, R109, R111, R113, R115, R117	R 0402_L	R	20
51k	贴片电阻	R47	R 0402_L	R	1
200K	贴片电阻	R48, R67	R 0402_L	R	2
100K	贴片电阻	R52, R78	R 0402_L	R	2
5.1K	贴片电阻	R54, R62	R 0402_L	R	2
120R	RES_0603	R61, R64	R 0603_L	RES_0603	2
100R	贴片电阻	R63	R 0402_L	R	1
75R	贴片电阻	R86, R90, R96, R99, R103, R107, R112, R116	R 0402_L	R	8
200R	RES_0603	R102, R104	R 0402_L	RES_0603	2
200Ω/0402*4/5%		RN1, RN2, RN3, RN4	RCA-8P4R-0402	RC-MT08W201JT	4
100Ω/0402*4/5%		RN5, RN6, RN7, RN8, RN9, RN10, RN11	RCA-8P4R-0402	RC-MT08W201JT	7
SW-PB3		S1, S2	SW-SMD_L4.0-W2.9-LS5.0	SW-PB3	2
H1102NLT		T1, T2, T3, T4	SMD_16	H1102NLT	4
0451010.MRL		U1	F2410	0451010.MRL	1
YT8512H		U2	QFN-32_L5.0-W5.0-P0.50-BL-EP3.4	YT8512C	1
ACM9070-701-2PL-TL01		U3	FILTER-SMD_4P-L9.0-W7.0-TL	ACM9070-701-2PL-TL01	1
TPS54302		U4, U8	SOT23-6L - duplicate	TPS54302DDCR_C311983	2
LM5013Q		U5, U7	ESOP8_N	LM5164DDAR	2
INA180A2DBVR	No Description Available	U6	SOT23-5	INA180A1IDBVR	1
CH217K		U9, U16	SOT23-6 - duplicate	CH217K	2
B0505S-1WR3		U10	PWRM-TH_4P-L11.5-W6.0-P2.54_B1503S	B0505S-1WR3_C7465178	1
ME6210A50M3G		U11, U13	SOT23-3L - duplicate	ME6210A50M3G	2
REF3033AIDBZR		U12	SOT23-3L - duplicate	REF3033AIDBZR	1
SX1308		U14	SOT23-6 - duplicate	SX1308	1
ACT45B-510-2P-TL003		U15, U18	FILTER-SMD_4P-L4.5-W3.2-BL_FACT4532	FACT4532-510-2P-T	2
ISO1044BDR		U17	SOP8_N	ISO1044BDR	1
MAX13487EESA+T		U19	SOP8_N	MAX13487EESA+T	1
RTL8305NBI-VB-CG/QFN-48		U20	QFN-48 L6*6	RTL8305NB-CG/QFN-48	1
SHT30-DIS-B10KS		U21	DFN-8_L2.5-W2.5-P0.50-BL-EP_1	SHT30-DIS-B10KS	1
STM32F407VET6	ARM Cortex-M4 32-bit MCU+FPU, 210 DMIPS, 512 kB Flash, 192 kB Internal RAM, 82 I/Os, 100-pin LQFP, -40 to 85 degC, Tray	U22	LQFP100_L	STM32F407VET6	1
AT24C256C-SSHL-T		U23	SOP-8	AT24C256C-SSHL-T	1
A-SP1943B1GHR6C-C01-2T		U24	LED-SMD_4P-L1.6-W1.5-BL_A-SP1943B1GHR6C-C01-2T	A-SP1943B1GHR6C-C01-2T	1
XT-3225-25MHz		X1, X2	OSC 3225-4P	XT-3225-25MHz	2
8MHz		X3	OSC 3225-4P	X32258MOB4SI	1